

Various Phases of Deployment an Application

- 1) Development Phase: Write and transfer to central server or use GitHub etc.**
- 2) Compilation: Compile the code to create a Build(Java Build, .NET Build etc.)**
- 3) Build Automation Tools: Apache Maven, Gradle, Ant, Bazel**
- 4) Testing: Junit, Nunit, Selenium, Jmeter, OWASP ZAP etc.**
- 5) Deployment(Stage Site & Live/Production Site)**
- 6) Monitoring and Logging Tools: Prometheus, Grafana, Datadog, Splunk**

What is CI/CD

- **Continuous Integration (CI):** Automate code integration to ensure that changes from multiple developers are regularly merged, tested, and validated.
- **Continuous Delivery (CD):** Ensure the codebase is always in a deployable state and automate the release process up to a staging environment. CD ensures that, once approved, the latest version of the software can go live with minimal manual effort.
- **Continuous Deployment (CD):** Take Continuous Delivery one step further by fully automating deployment to production after successful tests. With Continuous Deployment, there's no need for manual approval to release the code to production. Every change that passes tests is automatically deployed.

Key Benefits of CI/CD Across All Phases

Aspect	CI/CD Contribution
Speed	Automates repetitive tasks, accelerates deployment cycles, and reduces time-to-market.
Reliability	Automated testing, staging environments, and rollbacks make releases more predictable and stable.
Efficiency	Reduces manual work for developers and testers, allowing them to focus on high-value tasks.
Quality	Continuous testing and validation at every step improves the overall quality of releases.
Collaboration	Teams work more cohesively due to clear visibility of pipeline stages and faster feedback loops.
User Satisfaction	Frequent, stable releases lead to better and more timely features, improving the end-user experience.

Various Tools in CI/CD

- **Source Code Management (SCM):** GitHub, GitLab, Bitbucket
- **Continuous Integration (CI) Tools:** Jenkins, Travis CI, CircleCI, GitLab CI/CD
- **Containerization and Orchestration Tools:** Docker, Kubernetes
- **Continuous Deployment/Delivery (CD) Tools:** Spinnaker, Octopus Deploy, Argo CD, AWS CodeDeploy, Helm